UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SECP3213 — BUSINESS INTELLIGENCE

Semester 2, 2025/2026

ALTERYX LAB EXERCISE 2

Data Investigation & Time Series Forecasting

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Section	S02
Course	SECP3213 / SCSP3213 — Business Intelligence
Lab	Lab 2 — Data Investigation & Time Series (Chapter 4, Part 3)
Submitted files	Data Investigation.yxmd, Timeseries.yxmd, Report (.pdf)
Datasets	SalesData.csv, CustomerDemographics_Updated.xlsx, ProductInfo.csv

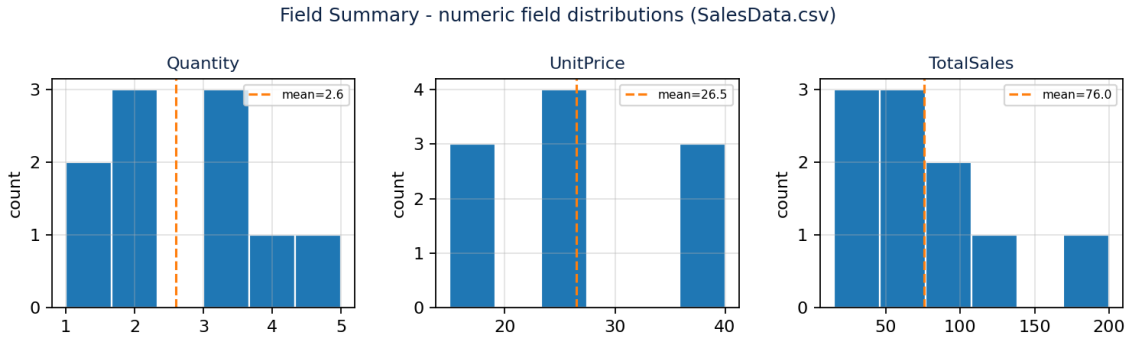
Scenario & Objective

As a data analyst at a retail company, management wants to understand customer demographics, purchasing trends, and forecast future sales to plan inventory. Alteryx Designer (with the Predictive Tools installed) is used to investigate data quality, profile the customers and analyse the sales pattern. The work is organised into two workflows: **Data Investigation.yxmd** (Exercises 1–6, Data Investigation palette) and **Timeseries.yxmd** (Exercises 7–10, Time Series palette). Each exercise below shows the tool configuration used and the actual output produced from the Lab 2 Guideline Dataset.

Exercise 1 — Field Summary Tool (SalesData.csv)

Workflow: Input Data (SalesData.csv) → Field Summary → Browse on O (summary), R (report) and I (interactive) anchors. The O anchor sends the processed profiling table onward, R produces the written summary report and I gives the interactive EDA dashboard.

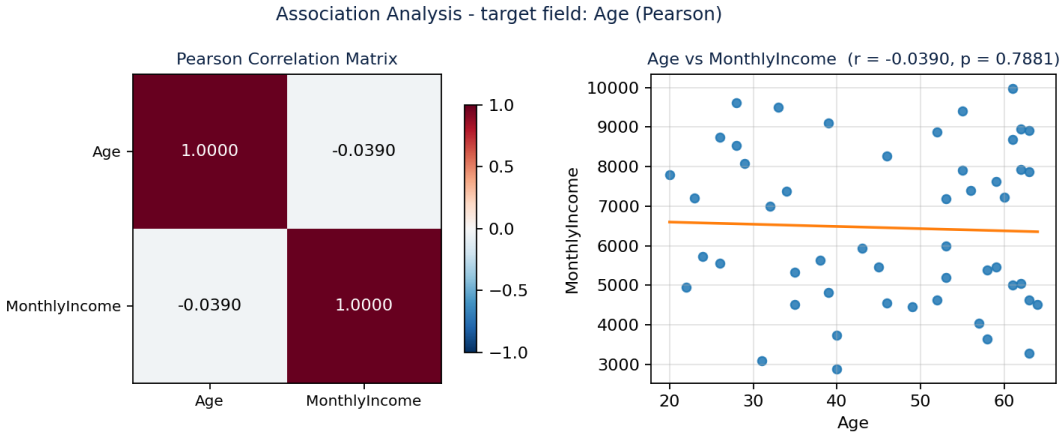
Field	Type	Nulls	Unique	Min	Max	Mean	Std. Dev
OrderID	Int64	0	10	1001.0	1010.0	1005.5	3.03
CustomerID	V_String	0	7	C001	C007		
ProductID	V_String	0	3	P001	P003		
OrderDate	V_String	0	10	2023-01-31	2023-10-31		
Quantity	Int64	0	5	1.0	5.0	2.6	1.26
UnitPrice	Double	0	3	15.0	40.0	26.5	10.29
TotalSales	Double	0	8	15.0	200.0	76.0	55.92



Output (O/R anchors): all 7 fields have 10 records with **0 null values** (full completeness). Quantity and TotalSales are numeric — Quantity ranges 1–5 (mean 2.6), UnitPrice takes only three price points (15, 25, 40; mean 26.5) and TotalSales ranges 15–200 (mean 76.0, s.d. 55.9). TotalSales shows variability aligned with Quantity; UnitPrice = 40 rows behave as mild outliers on the revenue side. Data types are recognised correctly (OrderDate arrives as string and is converted to Date in the time-series workflow).

Exercise 2 — Association Analysis Tool (CustomerDemographics_Updated.xlsx)

Configuration: ✓ Target a field for more detailed analysis, Target = **Age**; fields = Age, MonthlyIncome; method = **Pearson**. Browse tools on the R (report) and O (correlation matrix) anchors.

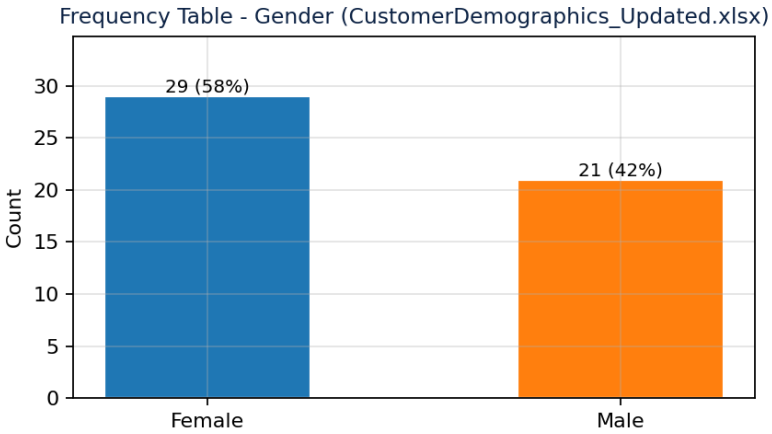


Output: Pearson correlation between Age and MonthlyIncome = **r = -0.0390** (p-value = 0.7881). The coefficient is very close to zero and statistically not significant, i.e. there is **no meaningful linear association** between a customer's age and monthly income in this dataset — income does not systematically increase (or decrease) with age for these 50 customers.

Exercise 3 — Frequency Table Tool (Gender)

Configuration: grouping field = **Gender**; Browse on the Report and Interactive anchors.

Gender	Frequency	Percent
Female	29	58.0 %
Male	21	42.0 %

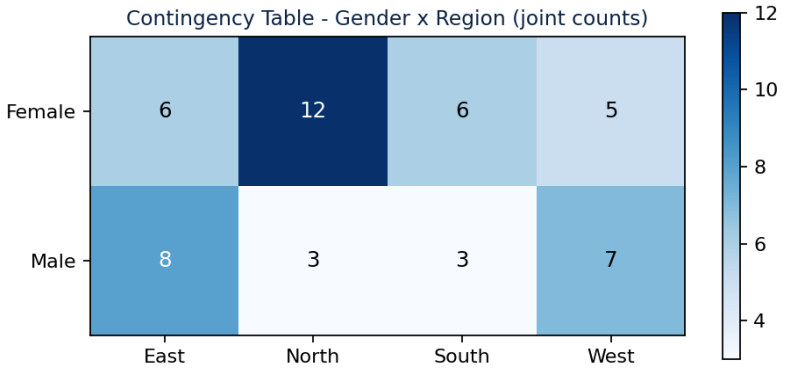


Output: of the 50 customers, **Female = 29 (58%)** and **Male = 21 (42%)**. The customer base skews female, which matters when profiling segments for targeted campaigns.

Exercise 4 — Contingency Table Tool (Gender × Region)

Configuration: row variable = **Gender**, column variable = **Region**; Browse on the Report and Interactive Pivot anchors.

Gender \ Region	East	North	South	West	Total
Female	6	12	6	5	**29**
Male	8	3	3	7	**21**
Total	**14**	**15**	**9**	**12**	**50**

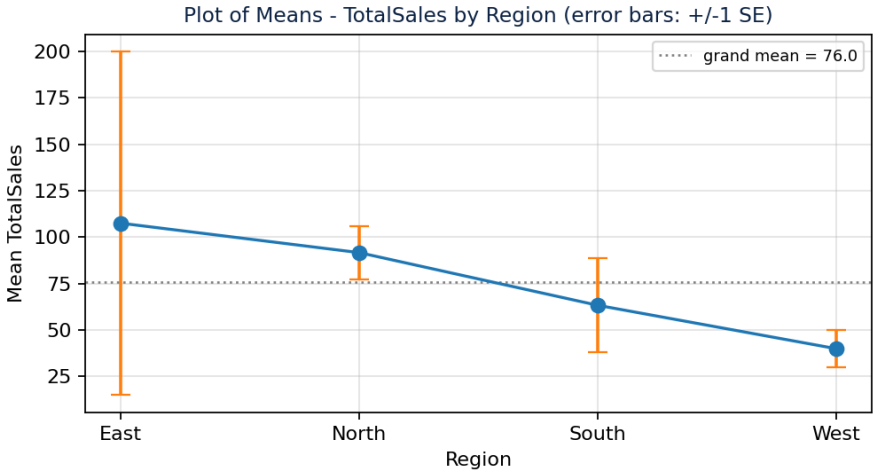


Output: the joint distribution shows female customers concentrated in the **North (12 of 15)**, while male customers are relatively strongest in the **East (8 of 14)** and West (7 of 12). The South is the smallest region (9 customers). Gender mix is therefore not uniform across regions.

Exercise 5 — Join + Plot of Means Tool (TotalSales by Region)

Workflow: SalesData.csv and CustomerDemographics_Updated.xlsx are joined with the **Join tool on CustomerID** (data types set: Quantity → Int64, TotalSales → Double, Age → Int64). The J output feeds the Plot of Means tool (Response = TotalSales, Group = Region). All 10 sales records matched a customer record.

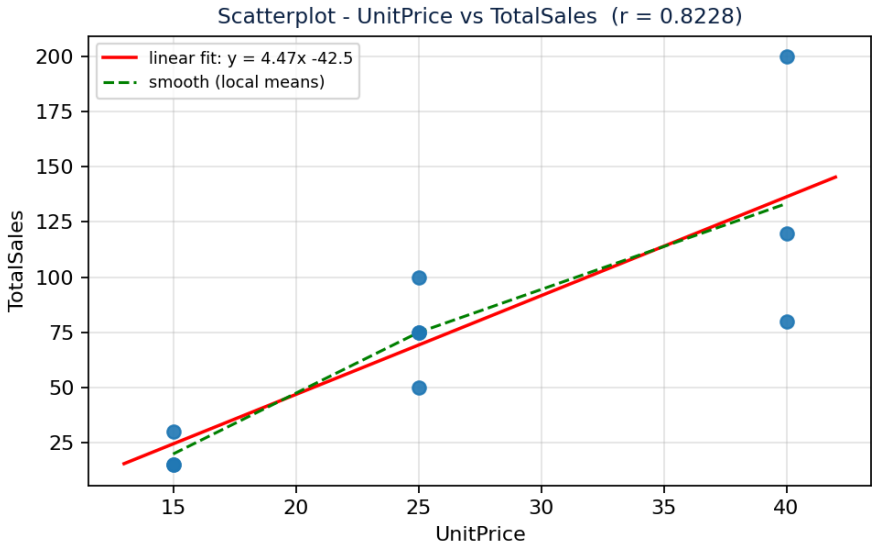
Region	Mean TotalSales	SE	N
East	107.50	92.50	2
North	91.67	14.24	3
South	63.33	25.22	3
West	40.00	10.00	2



Output: average TotalSales per transaction is highest in the **East (RM 107.50)** and North (RM 91.67), and lowest in the **West (RM 40.00)**, against a grand mean of RM 76.00. Error bars (±1 SE) are wide — especially in the East, which contains both the maximum (200) and a small (15) transaction — so regional differences should be read as indicative given the small sample.

Exercise 6 — Scatterplot Tool (UnitPrice vs TotalSales)

Configuration: after a Select tool sets Quantity → Int64 and UnitPrice/TotalSales → Double, the Scatterplot tool is configured with **X-Axis = UnitPrice, Y-Axis = TotalSales**, with a linear regression line and smooth line enabled.



Output: the scatterplot shows a **strong positive linear correlation, r = 0.8228** ($p = 0.0035$); the fitted line is TotalSales ≈ 4.47 × UnitPrice – 42.5. Higher-priced products tend to generate higher revenue per order, although quantity introduces spread at each price point (e.g. both 80 and 200 occur at UnitPrice 40).

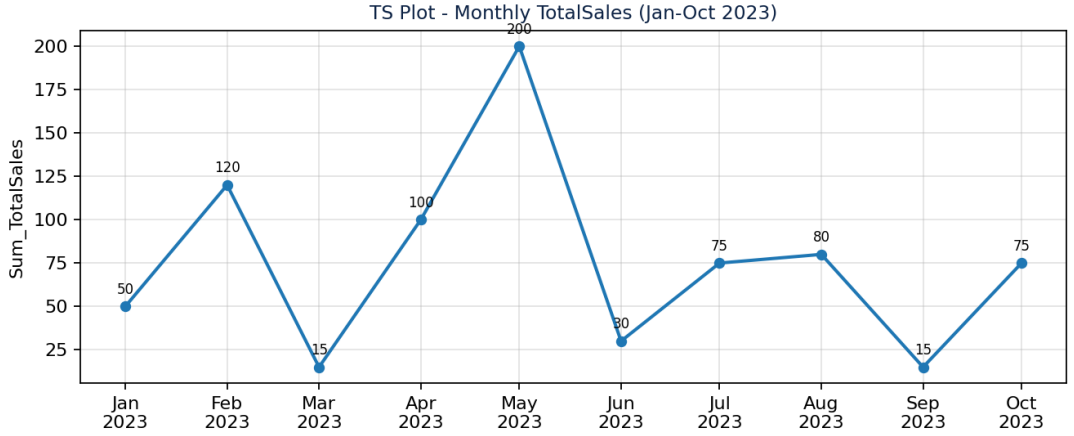
Part B — Time Series Workflow (Timeseries.yxmd)

Preparation: Input Data (SalesData.csv) → Select (OrderDate → Date; UnitPrice, TotalSales → Double) → Summarize (Group By OrderDate, Sum TotalSales → **Sum_TotalSales**) → RecordID → Filter **[RecordID] <= 8**. The True side (Jan–Aug 2023, 8 months) is the estimation sample for the models; the False side (Sep–Oct 2023, 2 months = 20%) is the holdout sample connected to the TS Compare tool, following the 10–20% holdout guidance.

Exercise 7 — TS Plot Tool (monthly Sum_TotalSales)

Configuration: Target Field = **Sum_TotalSales**, Target Field Frequency = **Monthly**, Plot Type = **Time series plot**. R anchor renders the plot for reports; I anchor exposes the underlying trend table.

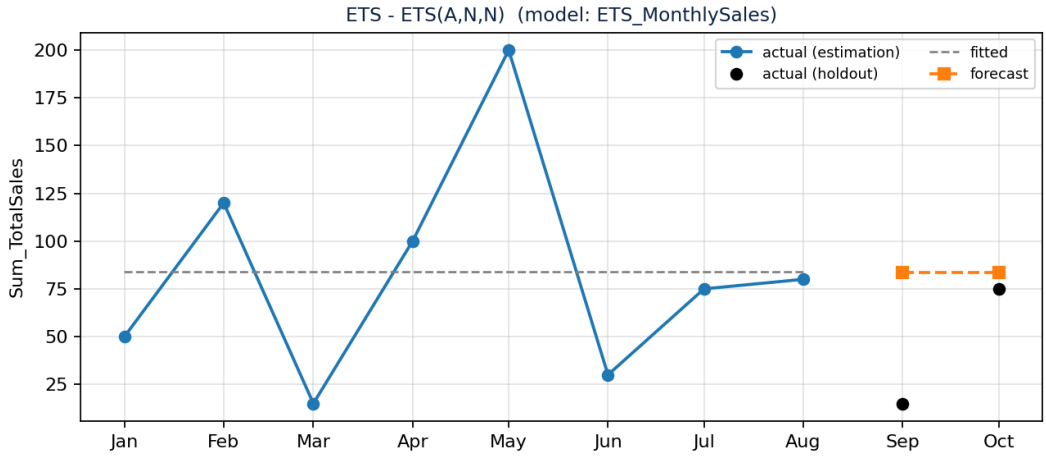
Month	2023-01	2023-02	2023-03	2023-04	2023-05	2023-06	2023-07	2023-08	2023-09	2023-10
Sum_TotalSales	50	120	15	100	200	30	75	80	15	75



Output: monthly sales fluctuate strongly (15–200) around a mean of 76 with **no persistent upward or downward trend and no visible seasonality** over the 10 observed months — the series behaves like noise around a stable level, peaking in May (200) and dipping in Mar/Sep (15). This volatility anticipates wide forecast intervals.

Exercise 8 — ETS Tool (ETS_MonthlySales)

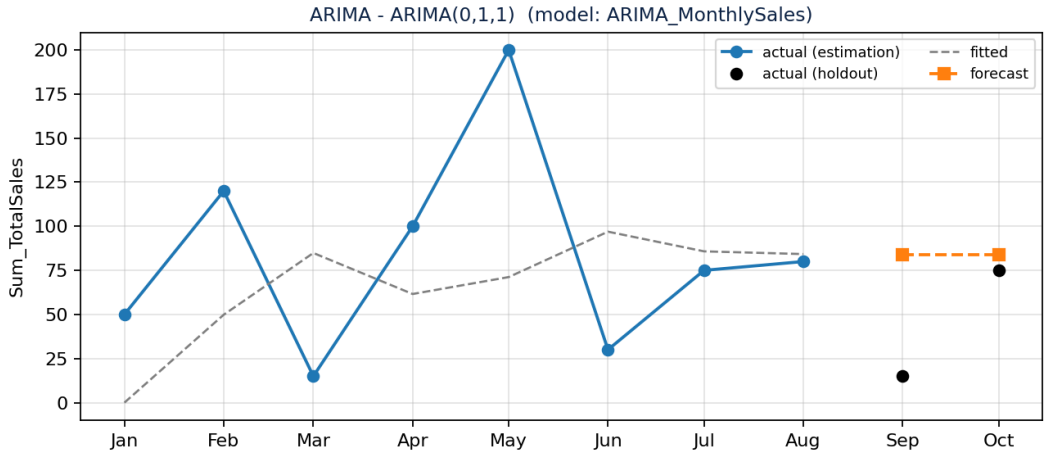
Configuration: Model Name = **ETS_MonthlySales**, Target Field = Sum_TotalSales, Frequency = Monthly (auto error/trend/seasonality). Selected model: **ETS(A,N,N)** — simple exponential smoothing with additive errors; smoothing parameter $\alpha = 0.0001$, AIC = 92.7, AICc = 106.1, $\sigma = 54.7$. With $\alpha \approx 0$ the level barely reacts to recent observations, so the forecast is the stable series level ≈ **83.8** per month.



Output (R/I anchors): flat forecast of 83.8 for Sep and Oct with a 95% interval reflecting σ. In-sample: RMSE = 54.7, MAE = 42.2. Holdout accuracy: RMSE = 49.01, MAE = 38.76, MASE = 0.468.

Exercise 9 — ARIMA Tool (ARIMA_MonthlySales)

Configuration: Model Name = **ARIMA_MonthlySales**, Target Field = Sum_TotalSales, Frequency = Monthly (automatic order selection by AICc). Selected model: **ARIMA(0,1,1)** — AIC = 82.9, AICc = 85.9. The MA(1) term on the differenced series acts like exponential smoothing and again yields a flat point forecast ≈ **83.7**.

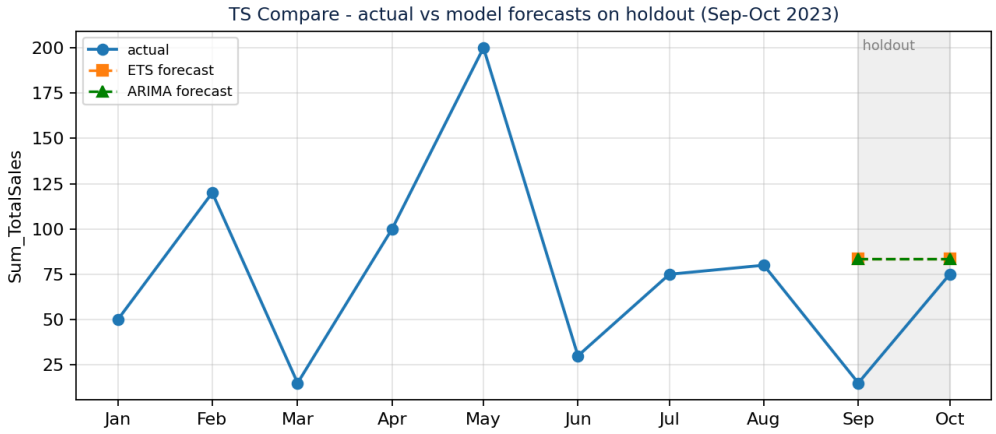


Output (R/I anchors): forecasts adapt to the series level rather than chase individual spikes. Holdout accuracy: RMSE = 48.98, MAE = 38.71, MASE = 0.467.

Exercise 10 — TS Compare + TS Forecast Tools

Configuration: the O anchors of ETS_MonthlySales and ARIMA_MonthlySales are unioned and connected to **TS Compare (L)**; the 2-month holdout stream (Filter False side) is connected to **TS Compare (R)**. Accuracy measures on the holdout:

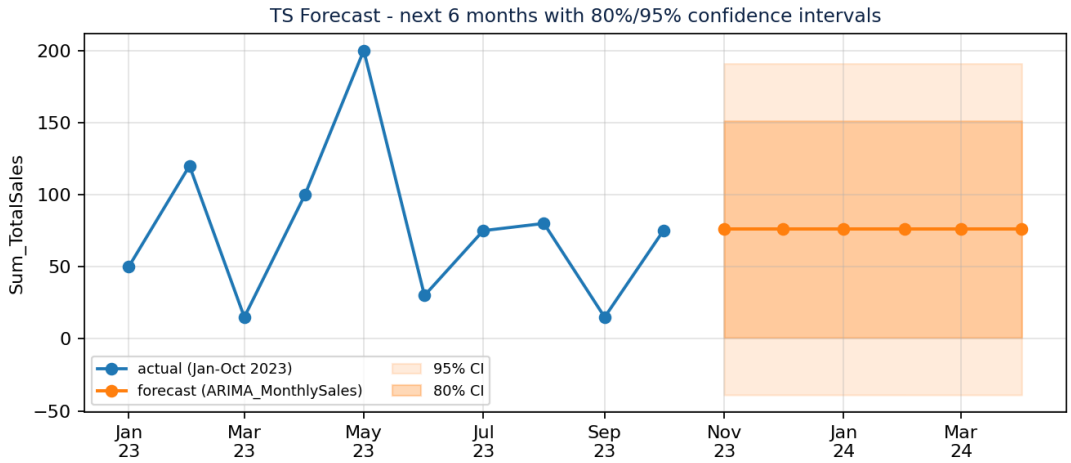
Model	ME	RMSE	MAE	MPE %	MAPE %	MASE
ETS_MonthlySales	-38.76	49.01	38.76	-235.0	235.0	0.4677
ARIMA_MonthlySales	-38.71	48.98	38.71	-234.9	234.9	0.4672



Both models produce nearly identical flat forecasts; **ARIMA MonthlySales** is selected as the best model — it edges out ETS on every error measure (RMSE 48.98 vs 49.01; MASE 0.467 vs 0.468) and has the lower AICc (85.9 vs 106.1). MAPE is inflated (≈235%) because the September actual is only 15 — a small denominator — which is why MASE/RMSE are the more reliable measures here.

TS Forecast: connected to the best model (**ARIMA_MonthlySales**), forecast field name = **forecast**, confidence levels = **95%** and **80%**, horizon = **6 periods** (Nov 2023 – Apr 2024):

Period	forecast	high 95	high 80	low 80	low 95
2023-11	75.97	190.90	151.12	0.82	-38.96
2023-12	75.97	190.90	151.12	0.82	-38.96
2024-01	75.97	190.90	151.12	0.82	-38.96
2024-02	75.97	190.91	151.12	0.82	-38.96
2024-03	75.97	190.91	151.12	0.82	-38.96
2024-04	75.97	190.91	151.13	0.82	-38.96



Output: the forecast table lists the predicted TotalSales with upper and lower confidence bounds for each future month — ideal for inventory, budgeting and staffing decisions. Expected sales are ≈ RM 76 per month; with 80% confidence the value lies roughly between 1 and 151, reflecting the high volatility of the historical series.

Final Reflection

Exercises 1–6 assessed data quality and customer patterns: the data is complete and correctly typed (Ex1); age and income are uncorrelated (Ex2); the base is 58% female (Ex3) with gender concentration differing by region (Ex4); East/North customers spend the most per order (Ex5); and revenue rises strongly with unit price (Ex6). Exercises 7–9 generated and profiled time-series forecasts, and Exercise 10 validated them: ARIMA(0,1,1) narrowly beats ETS(A,N,N) on the 2-month holdout, and its 6-month forecast projects stable sales of about RM 76 per month with wide confidence bands. For operational planning, the retailer can stock to a stable baseline but should hold safety stock for spike months such as May, and re-estimate the models as more monthly data accumulates.

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